

MEMORY CARD MATCHING GAME

Submitted By

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Academic Year: 2026

1. Introduction

The Memory Card Matching Game is a Python-based graphical application developed using the Tkinter library. The objective of the game is to find matching pairs of cards by flipping two cards at a time. This project helps in understanding Python programming concepts such as lists, buttons, event handling, and state management.

2. Objective

The main objective of this project is:

- To develop a simple interactive game using Python.
 - To understand GUI development using Tkinter.
 - To implement logical thinking and state management.
 - To improve programming and problem-solving skills.
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3. Tools and Technologies Used

- Programming Language: Python
 - GUI Library: Tkinter
 - IDE: Visual Studio Code
 - Version Control: GitHub
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4. Project Description

The game consists of pairs of cards placed face down. When a player clicks on a card, it flips and reveals its value. The player then selects another card.

- If both cards match, they remain visible.
- If they do not match, they flip back after a short delay.
- The game continues until all pairs are matched.

The application uses Python lists to store card values, buttons to represent cards, and state variables to track user selections.

5. Features

- Random card shuffling
- Interactive graphical interface
- Matching pair detection
- Win notification
- User-friendly gameplay
- State management for card tracking

6. Implementation

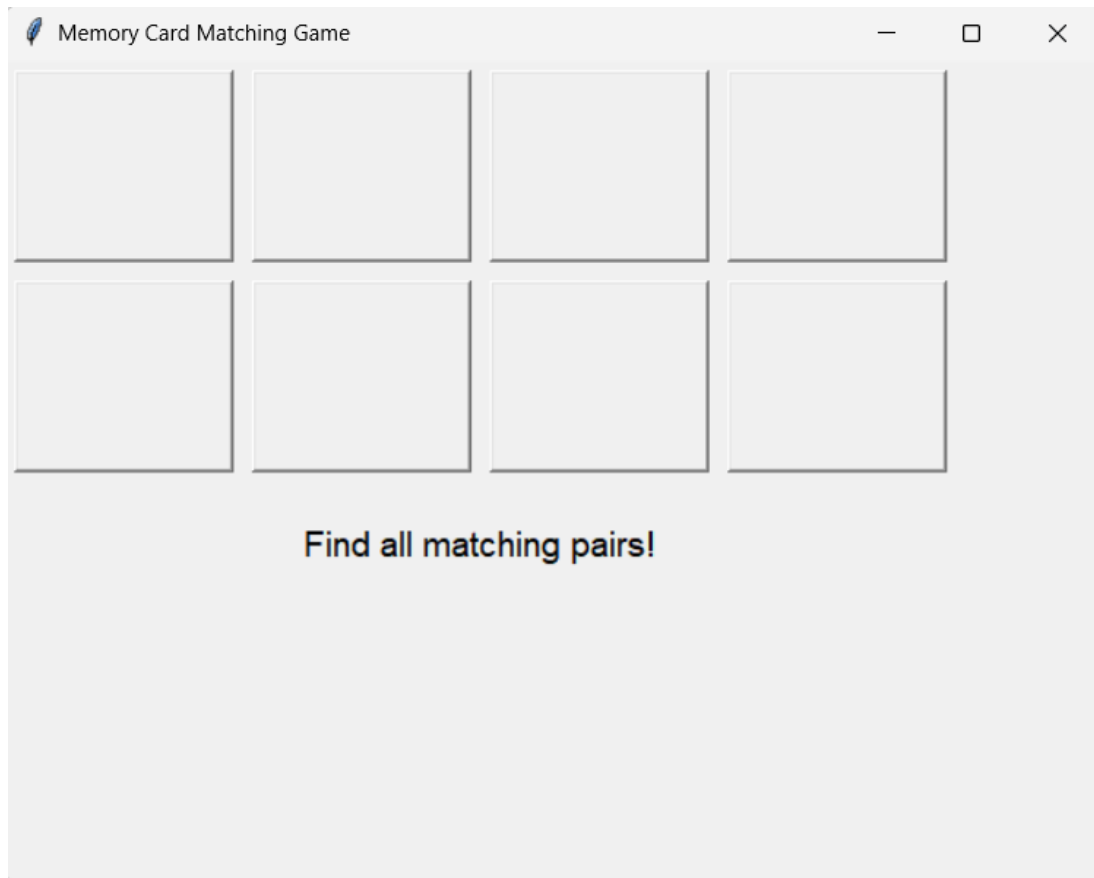
The project uses:

- Lists to store card values
- Random module to shuffle cards
- Tkinter buttons for card display
- Functions for game logic
- Variables to maintain game state

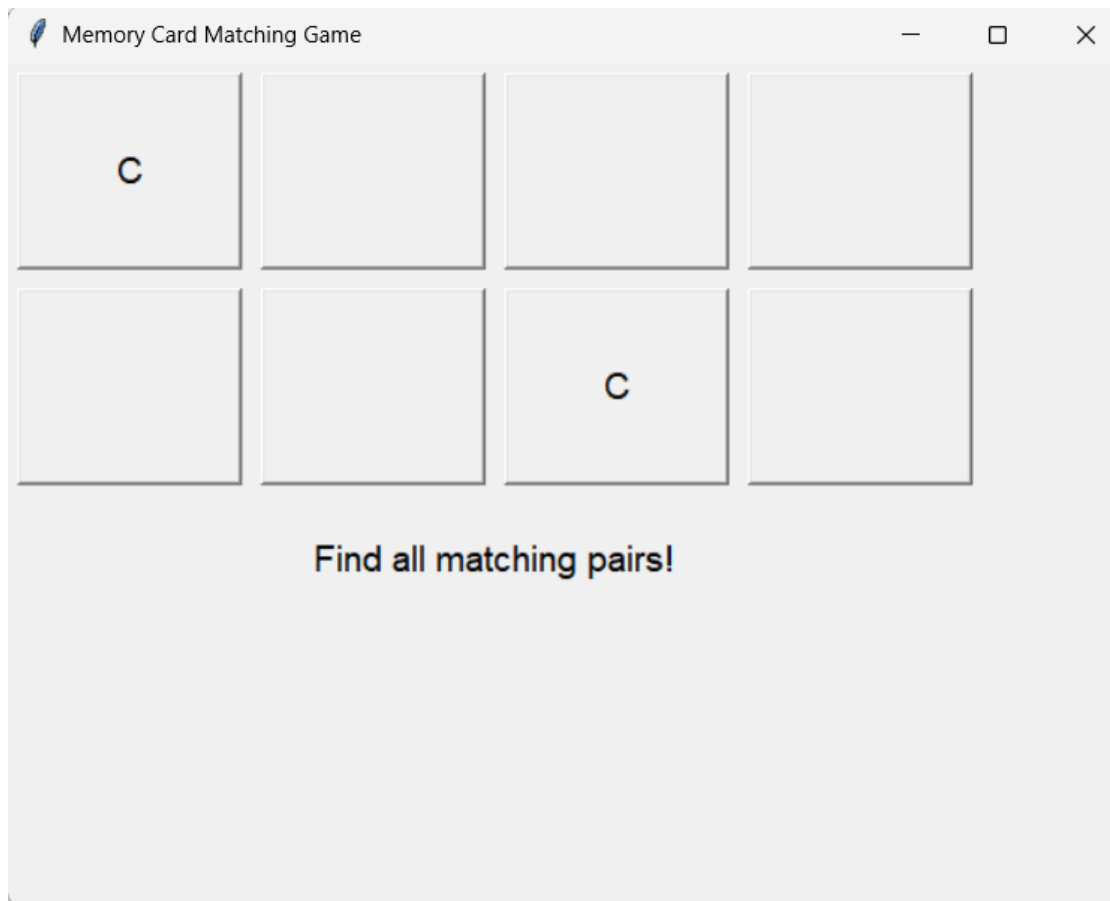
7. Screenshots

Insert the following screenshots here:

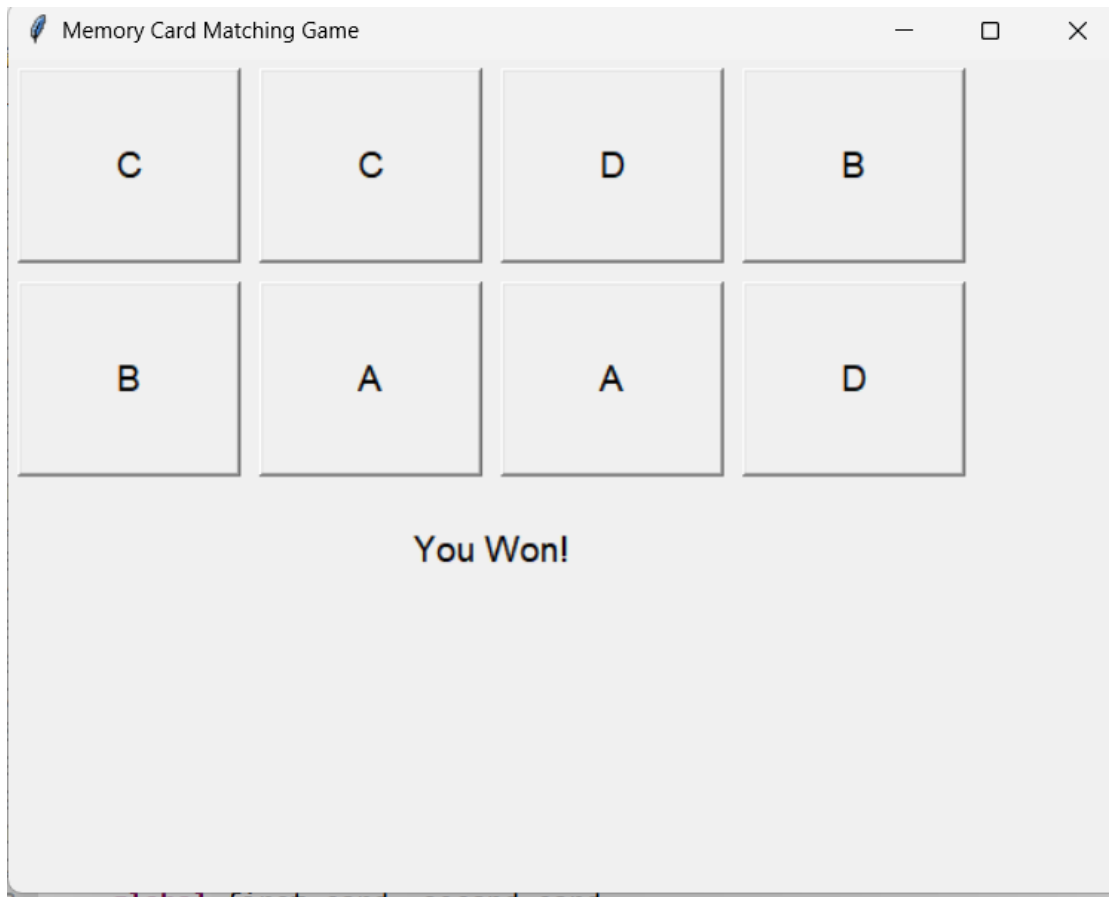
1. Game Start Screen



2. Gameplay Screen



3. Winning Screen



8. Learning Outcomes

Through this project, I learned:

- Python programming fundamentals
- GUI development with Tkinter
- Event handling techniques
- State management concepts
- GitHub repository management

9. Conclusion

The Memory Card Matching Game was successfully developed using Python and Tkinter. The project provided practical experience in GUI programming, logical thinking, and software development practices. It enhanced my understanding of Python concepts and GitHub project management.

10. GitHub Repository

Repository Link:

<https://github.com/ravix9145/memory-card-matching-game>